

Technology News

Use your voice to charge mobiles

Now you might never run out of your phone battery while talking, thanks to a new technique for turning sound into electricity that allows you to recharge phones using nothing but the power of the human voice.

Developed by electrical engineers at the Institute of Nanotechnology at Sungkyunkwan University in Seoul, the technology uses tiny strands of zinc oxide sandwiched between two electrodes. A sound-absorbing pad on top vibrates when hit by sound waves resulting in the tiny zinc oxide wires to compress and release. The movement generates an electrical current that can be utilised to charge a battery. The technology would also be able to employ background noise and music to charge the phone when it is not in use.

Log-in with your eye

New York-based Hoyos Group has designed a new security product that might allow you to log-in to your personal accounts with the blink of an eye.

EyeLock deploys iris-recognition as an alternative to passwords to log you into password-protected websites and applications. The device is the size of a standard business card and connects to the user's computer by a USB cable.

"Every time you log in, it reads your iris and creates a unique key which is a series of numbers, and this key changes every time you log in, so no one can hack it," said Tracy Hoyos, assistant marketing director, Hoyos Group.

Smartphone to predict traffic jams

Now you might get lucky and stay away from gridlock, thanks to a tech-



Smartphone predicting the traffic jam ahead

nology developed by IBM in collaboration with several universities and civic administration in California which would be able to predict traffic jams 40 minutes before they actually happen.

The smartphone application has location-sensing capabilities which automatically track the driver. The information is fed through the Internet to computers that identify patterns such as commutes to and from work. It then compares real-time traffic data with previous traffic patterns and sends the information about probable traffic trouble via text messages or e-mails to commuters taking regular routes.

The service is powered by a first-of-its-kind learning and predictive analytics tool called the Traffic Prediction Tool developed by IBM Research.

Microscope that detects skin cancer

Researchers at Germany's Fraunhofer Institute for Applied Optics and Precision Engineering IOF have developed a

microscope which is slim enough to fit in the palm of your hand, yet powerful enough to deliver images at a scanner-like resolution of five micrometres, over a wide surface area.

"Our ultra-thin microscope consists of not just one but a multitude of tiny imaging channels, with lots of tiny lenses arrayed alongside one another. Each channel records a tiny segment of the object at the same size for a 1:1 image," said Dr Frank Wipperfurth, who managed the team.

Each slice is roughly 300x300 μm^2 in size and fits alongside the neighbouring slice. A computer program then assembles these to generate the overall picture. The difference between this technology and a scanner microscope is that all of the image slices are recorded simultaneously.

Capable of a 36x24mm² image size, the microscope can capture matchbox-sized objects in a single pass. The spectrum of its applications is diverse. The microscope could help doctors scan patients for skin cancer more easily and